(iii) When the interest is compounded quarterly, we have

$$A = P\left(1+\frac{r}{400}\right)^{4n}$$

(iv) When the interest is compounded annually but rate of interest differs year by year, we have

$$A = P\left(1+\frac{a}{100}\right)\left(1+\frac{b}{100}\right)\left(1+\frac{c}{100}\right)\ldots$$

where a, b and c are interest rates for I, II and III years respectively.

(v) When interest is compounded annually but time period is in fraction say $a\frac{b}{c}$ years, we have

$$A = P\left(1+\frac{r}{100}\right)^{a}\left(1+\frac{\frac{b}{c}\times r}{100}\right)$$

(Use of calculators are permitted for lengthy calculations and also to verify answers).

Example 4.14

Find the C.I for the data given below:

(i) Principal = ₹4000, r = 5% p.a, n = 2 years, interest compounded annually.

(ii) Principal = ₹5000, r = 4% p.a, n = $1\frac{1}{2}$ years, interest compounded half-yearly.

(iii) Principal = ₹30000, r = 7% for I year, r = 8% for II year, compounded annually.

(iv) Principal = ₹10000, r = 8% p.a, n = $2\frac{3}{4}$ years, interest compounded yearly.

Solution:

(i) Amount, $A = P\left(1+\frac{r}{100}\right)^{n}$

$$= 4000\left(1+\frac{5}{100}\right)^{2}$$

$$= 4000 \times \frac{21}{20} \times \frac{21}{20}$$

$$A = ₹4410$$

$$\therefore C.I = A - P = 4410 - 4000 = ₹410$$

(ii) Amount, $A = P\left(1+\frac{r}{200}\right)^{2n} = 5000\left(1+\frac{4}{200}\right)^{2\times\frac{3}{2}} = 5000 \times \frac{51}{50} \times \frac{51}{50} \times \frac{51}{50}$

$$= 51 \times 10.2 \times 10.2$$

$$= ₹5306.04$$

$$\therefore C.I = A - P = 5306.04 - 5000$$

$$= ₹306.04$$

04_8th_Maths_Life Mathematics_Chap 4.indd 135 20/11/2022 13:13:15